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Can the Eurozone Survive?

In November 2009 a newly elected Greek government revealed that previous data had gravely understated the budget deficit. Greece sank into crisis as yields on its bonds sailed steadily upward—past 30% for ten-year maturities.[1] Despite cutting its primary deficit 4.2% per year from 2009 through 2011, arguably the most severe reduction in the developed world in recent decades, Greece effectively went bankrupt, and in early 2012 its lenders were forced to take a €100 billion loss on its debt.[2]

Though Greece produced just 2.5% of Eurozone GDP, its crisis ignited crises across southern Europe—now flaring, now subsiding, now flaring again—amid fears of the euro's collapse.[3] European leaders repeatedly vowed to do "whatever it takes" to save the euro, the *Financial Times* lamented, but "just as often their subsequent actions, or lack of them, have belied these fine words."[4]

In early 2012, things started looking up. After the European Union (EU) had promised a permanent €500 billion fund to lend to threatened governments and proposed a tough fiscal pact, the European Central Bank (ECB) loosed a "wall of money"—€2 trillion of loans to national banks in hopes that they would, in turn, lend to governments and industry.

But by summer, euro optimism was sinking again. Amid widespread talk of "Grexit," a Greek exit from the euro, Citi's chief economist went so far as to put a specific figure on the speculation. He said the probability that Grexit would occur within 12 to 18 months was 90%.[5] Meanwhile, yields on 10-year Spanish bonds reached 7.6% and those on Italian bonds 6.6%—levels that might well lead both nations into debt traps like Greece.[6]

Then, on the eve of the summer Olympics, ECB President Mario Draghi, reiterating a phrase that had been heard so often before, announced, "Within our mandate, the ECB is ready to do whatever it takes to preserve the euro. And believe me, it will be enough."[7] He seemingly intended to buy large quantities of bonds of troubled governments. If that was his idea, no one was sure he would get it past adamant opposition from the powerful German central bank, the Bundesbank. Still, yields on Spanish and Italian bonds dropped significantly, and European stocks surged 4% that day.

Why did contagion from tiny Greece so rapidly infect the entire Eurozone? Would the accretion of measures promised by the European Union (EU) and the ECB, in coordination with the International Monetary Fund (IMF), finally be adequate to staunch the crisis? And would the euro survive?

Senior Lecturer Dante Roscini and Research Associate Jonathan Schlefer prepared the original version of this case, "Can the Eurozone Survive," HBS No. 712-058, which is being replaced by this version prepared by the same authors. This case was developed from published sources. HBS cases are developed solely as the basis for class discussion. Cases are not intended to serve as endorsements, sources of primary data, or illustrations of effective or ineffective management.

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Creating the Euro

The international financial order negotiated in Bretton Woods, New Hampshire, in 1944, fixing the dollar against gold and other currencies against the dollar while allowing devaluations in exceptional circumstances, proved workable through the 1960s. But it fell apart when the United States devalued the dollar in 1971 and finally abandoned any fixed exchange rate in 1973.

The resulting exchange-rate volatility, two oil crises, and other troubles in the 1970s led to relatively high and erratic inflation. Exchange-rate volatility posed particular problems for nations of the European Community (EC), predecessor of the European Union, as EC members lowered mutual trade barriers. For example, when currencies fluctuate, firms face more uncertainty about what imported input materials will cost them or what their products will sell for abroad. As a result, they are likely to reduce investment.[8]

In 1979 the EC tried to stabilize currencies in a system called the Exchange Rate Mechanism (ERM). French and Italian politicians hoped the ERM would lower their rates of inflation by tying their currencies to the German mark.[9] German Chancellor Helmut Schmidt had an essentially political goal: "binding West Germany closer to Western Europe."[10] Despite the ERM, exchange-rates continued to fluctuate in the early 1980s but began to settle down late in the decade.

After the Single European Act, entering into force in 1986, called for eliminating all barriers to trade, financial flows, and migration within the EC, many saw a single currency as the natural next step. Some argued that makeshift bands like the ERM were bound to fall apart sooner or later; the only viable systems, in this view, were fully flexible exchange rates or a single currency.[11]

The EC planned a single currency. Again, the goals were political as much as economic. Having fought repeated wars with Germany, France feared its unification after the Berlin Wall fell in 1989. French President François Mitterrand saw a common currency as cementing a more peaceful Germany and agreed to back the united Germany's entry into the EC (in effect, it was a new country) if German Chancellor Helmut Kohl backed a common currency.[12] Kohl believed a common currency would provide a core of European stability as former members of the Eastern Bloc joined it.[13] And even he said it would bring peace by "containing a potentially dangerous Germany within Europe."[14] Not everyone agreed. The U.S. economist Martin Feldstein wrote, "Germany's assertion that it needs to be contained in a larger European political entity is itself a warning. Would such a structure contain Germany, or tempt it to exercise hegemonic leadership?"[15]

The so-called Maastricht Treaty forming the European Union (EU) and laying the road to monetary union was signed in 1992. That very year, the ERM came under attack again. Finland, Sweden, and Norway stopped trying to peg their currencies, while Spain, Portugal, and Ireland had to devalue, Italy was forced to let the lira float, and the British pound crashed. But by July 1, 1998, eleven states were deemed to have met the "convergence" criteria for adopting the single currency, the euro: among other things, maintaining fiscal deficits under 3% of GDP and limiting government debt to 60% of GDP. They adopted the euro on January 1, 1999.

In fact, six of the eleven original Eurozone members did not meet the debt limits (**Exhibits 1 and 2**). However, a weighted majority vote of member governments had the actual power to decide which nations adopted the euro, and political criteria prevailed.[16] A Stability and Growth Pact (SGP) requiring members that violated debt and deficit limits to be monitored under an "excessive deficit procedure" was often ignored. In the first decade of the euro, Germany violated the debt limits every year but one, and France violated them six years. Perhaps worst, in 2003, when Germany and France exceeded the deficit limits, the European Commission, the EU's administrative body, called for an

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excessive deficit procedure, but the heads of state meeting in the Council of Ministers, where Germany and France held heavily weighted votes, brushed the commission aside.[17] This was a turning point, says Philip Maystadt, former Belgian finance minister.[18] Henceforth, other finance ministers concluded they would not have to pay much attention to the SGP either.

The Euro: More Complicated Than It Looks

The euro was an unprecedented experiment. No other major nation lacked its own currency.[19] The Treaty on European Union, deeply held beliefs, and the structure of national economies would profoundly influence this experiment. These factors help explain why the Eurozone, with a 6% of GDP deficit and 85% of GDP debt in 2010, was threatened with sovereign crises, while the United States, with substantially higher deficits and debt, was not (**Exhibits 1 and 2**). Similarly, Spain, a Eurozone member with a 9% of GDP deficit and 61% of GDP debt, entered into crisis in 2010, while Britain, still on the pound sterling, with a 10% of GDP deficit and 80% of GDP debt, did not.

Article 125 of the Treaty on European Union, the "no-bailout clause," forbade the European Union or national governments from assuming the debt of other member governments.[20] After a government borrowed 60% of GDP, the treaty demanded budget cuts and tax increases, even if they deepened a recession. "If you live beyond your means, then you can repair your balance sheet only if your consumption goes down," said Jörg Krämer, chief economist of Commerzbank in Frankfurt.[21]

A driving force behind debt and deficit limits, particularly after the crisis struck, was German inflationary fear. Hyperinflation—inflation so high that bread prices change literally by the hour and life savings evaporate—had devastated Germany after World War I and contributed to the downfall of its first democracy, the Weimar Republic. Memories of that hyperinflation permeated the German mentality as only the most cataclysmic national experiences can.

Most advanced nations such as the United States could deploy expansionary fiscal policies during a recession. Moreover, the U.S. government contributed vastly more to its states' finances than the European Union did to its members' finances. The U.S. government paid for, among other things, social security, Medicare for those over 65, Medicaid for the poor, and housing subsidies; the military; much of the infrastructure such as highways; and some education, police, and other primarily state responsibilities. When a U.S. state fell into recession, for every dollar of per capita income that residents lost, federal taxes were reduced by 34 cents and federal transfers to the state increased by 6 cents.[22] In the European Union such tax reductions and transfers amounted to practically zero.

The Treaty on European Union stated that the "primary objective" of the European Central Bank (ECB) was to "maintain price stability."[23] By contrast, U.S. Employment Act of 1946 enjoined the government, including the Fed, to promote full employment, and the Bank of England traditionally could do the same.[24] Whether they succeeded was another question, but they had the mandate.

The ECB was prohibited from directly lending to governments or buying their bonds. [25] In legal terms, the Fed or Bank of England were similarly limited. But these legal prohibitions had little practical effect. If governments sold bonds to private investors on Monday, central banks could buy them on the secondary market on Tuesday. In fact, during the euro crisis, the ECB did so.

But the culture of the ECB had enormous practical effect. While the Fed was created with the primary goal of staunching financial crashes such as had plagued the nineteenth- and early twentieth-century U.S. economy, the ECB was created with the primary goal of preventing inflation such as had plagued many nations in the 1970s. During the 2008 financial crisis, the Fed not only

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loaned hundreds of billions of dollars to troubled banks but deployed an alphabet soup of programs to purchase trillions of dollars' worth of troubled financial instruments.[26] It bought mortgages and commercial paper; it channeled loans through financial institutions to consumers and businesses, large and small. By contrast, the ECB had a weak mandate to act as a lender of last resort; buying sovereign bonds particularly troubled officials.[27]

The structure of Eurozone banking made it vulnerable to crisis. Total European bank assets were large, generally between 200% and 400% of national GDP, while U.S. bank assets were less than 100% of GDP.[28] As well, national Eurozone governments, responsible for rescuing troubled banks, lacked authority to print euros to rescue them.[29] These governments had to take on euro debt if they needed resources to rescue national banks. By contrast, the Fed, responsible for rescuing major troubled banks along with other U.S. agencies, could, of course, print unlimited dollars.[30]

The close relationship between Eurozone governments and banks meant that sovereign-debt crisis would be particularly perilous. Because of historical practice, national banks held large portions of their governments' debt. Banking regulations encouraged this practice by counting sovereign debt as risk-free.[31] In 2011, Greek, Irish, Portuguese, Spanish, and Italian banks held 20% to 30% of their own governments' debt, while U.S. banks held 2% of U.S. debt (see **Exhibit 4**). A sovereign debt crisis therefore could build into a vicious cycle, as the government would have to provide funds to rescue banks, but banks would have to continue lending to the government to roll over its debt.

Further doubts about the euro concerned the extent to which Europeans saw themselves as European rather than German, Italian, or Greek. For all their conflicts, Americans were Americans; nobody even noticed when resources passed from, say, New York to Alabama. U.S. coins had images of founding fathers. Michel Prieur, a French numismatist, noted that euro coins and bills lacked any unifying symbols and depicted only "bridges that come from nowhere and that lead nowhere."[32]

Germany

During the crisis, as EU officials urged troubled nations to spur growth via "structural" reforms such as making labor markets "flexible" (a term usually left vague) or lightening business regulation, they often pointed to the German model. In particular, Chancellor Angela Merkel and others held up German labor-market reforms of the early 2000s as key to sustaining growth and employment.[33]

In the 1990s, Germany had earned a reputation as the "sick man of Europe." The reasons could be debated. It faced intensified competition from Asian and Central European manufacturers, and the former West Germany had absorbed the economically troubled East Germany. Unemployment in eastern Germany was 14% in 2001, while in western Germany it was only 6%.[34] Whatever the causes, many Germans, economists, and reporters blamed unions.[35]

A strike in June 2003 by IG Metall, a powerful union spanning a range of industries, aroused broad dissatisfaction and ended in defeat.[36] This defeat helped Chancellor Gerhard Schröder pass a package of labor reforms. They were seen as modest. "'It's not a great reform package," said Rudolf Hickel, an economist at the University of Bremen. "But we're muddling through."[37]

The reforms hardly crippled German unions. The Organisation for Economic Co-operation and Development (OECD), a group of mostly advanced nations, assesses economic characteristics of its member states (**Exhibits 5 and 6**). If "flexible" labor markets are defined as weak protection for workers—few obstacles to dismissal and low unemployment benefits—they certainly did not characterize Germany. Compared with several other Eurozone states, as well as the United States and

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Sweden, Germany had the strongest protection against dismissal for "regular," non-temporary, employees, except for Portugal. It had the most generous long-term insurance and other assistance for the unemployed, except for Sweden and Ireland.

The Schröder reforms did decentralize labor-relations, argues the political scientist Richard Deeg.[38] Individual firms and councils representing workers increasingly bypassed industry-wide contracts to tailor pacts to their needs. For example, in one type of pact, when a firm was pressed by competition, workers would accept wages below the contract level in exchange for firms' promise to invest in Germany.[39] Such firm-level pacts arguably improved German industrial performance. Thus, concludes Deeg, "One of the great ironies . . . is that despite all the calls for radical change in the labor relations system, it has remained broadly stable while exhibiting notable flexibility."[40]

During both the global financial crisis and the euro sovereign-debt crisis, German firms rarely resorted to laying off workers, as the labor "flexibility" slogan might imply, but reduced working hours per employee.[41] Transfers from the government—the famous German social safety net—helped soften pay losses. Health, pension, and unemployment insurance benefits were unaffected.

Spain and Italy

The peripheral Eurozone nations—Greece, Ireland, Portugal, Spain and Italy—shared commonalities. All ran current-account deficits, borrowing from abroad, between 2000 and 2008. The Greek deficit averaged 9% of GDP; Spain's averaged 6% of GDP (**Exhibit 8**). Italy's was more moderate, its current-account balance in 2000 declining to a 3% of GDP deficit in 2008.

However, the underpinnings of these current-account deficits differed sharply. Everyone pointed to the Greek case: the government spent too much, collected too little tax, ran excessive deficits, and borrowed from abroad to cover the shortfall. But the Spanish case was just the opposite. The government ran budget surpluses as its debt fell from 59% of GDP in 2000 to 40% in 2008 (**Exhibit 2**). Massive *private* sector borrowing drove current-account deficits. Italy ran primary fiscal surpluses but small overall deficits (counting interest payments).

Because of their size, Spain and Italy posed the gravest threats to the euro when crisis erupted. Spain's troubles stemmed largely from a housing bubble. With little more than 40 million inhabitants in 2000, it received nearly 6 million immigrants through 2008, largely from Latin America, Eastern Europe, and Africa.[42] Many retirees arrived from northern Europe, and a rise in divorces created new households.[43] All these groups took out mortgages and built homes. Capital flowing in from abroad, principally from core European banks, supplied the funds for the mortgages.

Spanish productivity growth from 2000 through 2008 was a poor 0.7% per year.[44] A possible culprit was the labor relations system. Like other peripheral Eurozone nations, Spain had a dual market. Workers on long-term contracts were costly to lay off: firms had to pay 45 days salary per year on the job. Workers on short-term contracts of up to a year could be laid off cheaply: firms only had to pay 8 days of salary or less. Younger workers thus constituted a contingent labor force hired and fired at will, so firms had little incentive to invest in training them.[45] Just the group that should have been upgrading its skills and helping raise productivity was instead marginalized.

When the global financial crisis hit, the Spanish government deployed expansionary policies, providing a €400 credit to each tax payer and €2,500 per newborn. The budget went from a 1.9% surplus to a 4.5% deficit (**Exhibit 1**). Despite these efforts, after growing 3.5% in 2007, the economy barely grew in 2008 and contracted 3.7% in 2009 (**Exhibit 9**). Official unemployment reached 20% in

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2010 (**Exhibit 10**).[46] Part of the problem was the collapse of the construction sector that had propelled growth during the housing bubble. Employment in that sector sank from 2.7 million in 2007 to 1.6 million in 2010, and output fell 15%.[47] Fiscal stimulus could not revive it.

Italy posed an even larger concern than Spain. After Italian public debt had surged to over 120% of GDP in the early 1990s, Prime Minister Giuliano Amato, considered a technocrat, launched a fiscal austerity effort.[48] The government pushed the primary surplus to between 4% and 6% of GDP and continued to run a primary surplus through 2008. Yet government debt, having topped out at 122% of GDP in 1994, fell only to 103% of GDP by 2007, and then during the crises it shot right back up to 120% of GDP (see **Exhibit 2**).

The key problem was poor growth. A growing economy lowers debt as a percent of GDP. But from 1990 through 2010, as the U.S. growth averaged 2.5% a year and the EU growth 1.7%, Italian growth averaged only 1% (see **Exhibit 9**).[49] Italy's productivity growth was just 0.1% per year from 2000 through 2008.[50] As with Spain, the labor-relations system might have been a cause. Italy did not have standard unemployment insurance that compensates workers who lose their job. Instead, if a firm laid off employees, it could secure income protection for them from the government, and when it was able to expand production again, it hired them back. But to participate in this program, workers were barred from moving. They had little incentive to search for another job.

Italy also had a dual labor market. The infamous Article 18 of the labor code, passed in the 1970s, prevented firms of more than 15 employees from laying off workers except for gross misconduct.[51] A firm had to prove misconduct in court and, if unsuccessful, had to pay back wages and legal costs. Precisely to instill more "flexibility," the government established short-term contracts allowing younger workers to be laid off easily.[52] The results seemed counterproductive, like those in Spain. And small firms resisted hiring more than 15 workers, even if it could gain economies of scale.

The Eurozone Periphery Syndrome

During the decades when Italy, Spain, Greece, and Portugal had their own currencies, with inflation levels higher than Germany's, they were periodically forced to devalue against the mark. Sensibly fearing further devaluation, lenders charged borrowers in those nations high interest rates. In 1995, benchmark 10-year government bonds of Italy, Spain, and Portugal yielded 4% to 6% more than comparable German bonds, while those of Greece yielded 12% more.[53]

As these nations were believed to be realigning their economic policies to adopt the euro, confidence soared. They would devalue no more. Yet their interest rates remained relatively high, so banks in Germany and other nations saw profitable opportunities and poured loans into the periphery. Spreads over German bonds dramatically declined (see **Exhibit 11**).

If those loans had been invested in productive industries, all might have been well, but they were not. Capital flows into Spain, channeled through the private sector, drove a housing bubble. Capital flows into Greece, channeled through the public sector, allowed government salaries and benefits to support consumption. Capital flows into Italy supported neither blatant government profligacy nor an obvious asset bubble, but did keep a stagnant economy afloat.

Capital flows, by inflating periphery economies, allowed workers to boost wages. From 2000 to 2008, labor costs rose 15% in Germany but 28% in Italy and 42% in Spain.[54] With fast-rising wages and stagnant productivity growth, Italy and Spain saw unit labor costs (the hourly labor cost to produce an estimated constant quantity of output) rise more than 20%, while Germany, with slow-

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rising wages and better productivity growth, saw flat unit labor costs (**Exhibit 12**). The periphery thus lost some 20% of its competitiveness against Germany.[55] In other words, it lost the potential to export to repay loans.

Capital inflows prepared the way for other problems as they passed through the banks, temporarily boosting their profits but making their balance sheets more precarious. For example, when northern Europeans bought homes in Spain, their payments were deposited in Spanish banks; when German institutions bought Italian government bonds, their payments found their way into Italian banks.[56] Peripheral banks' deposits thus increased. The banks also took on other liabilities such as by selling bonds to foreign banks. Of course, the peripheral banks used the funds thus obtained to increase their assets, for example, by providing mortgages or buying government bonds, that paid high returns.

The banks thus increased their leverage, a common process during a boom. It works this way: "Tier 1" capital is bank equity plus retained profits.[57] It increases as profits rise in a boom. But being optimistic about future profits, banks often borrow (or otherwise increase liabilities) and make loans (or otherwise increase assets) even faster than their capital increases. Leverage, the ratio of assets to capital, rises. Bank leverage rose in many countries in the early 2000s, but Eurozone banks became especially leveraged, with assets reaching 35 times capital (**Exhibit 3**).

Leverage boosts profits while things go well but causes trouble if things go badly. Accounting for losses on bad or risky loans depends on applicable regulations, but those losses must be deducted from capital.[58] If a bank is leveraged 33 to 1—if its loan portfolio is 33 times its capital—an approximately 3% loss on its loans completely wipes out its capital, making it insolvent.

A Crisis Ignites

In early 2010 the new Greek government began to cut spending and raise taxes. European leaders pledged "determined and coordinated action" to support the euro but said Greece must clean up its own finances.[59] U.S. Treasury Secretary Timothy Geithner and Fed Chairman Ben Bernanke, urging decisive EU action in April, were stunned to hear that EU leaders proposed only a €60 billion loan fund.[60] As Greek bonds were downgraded to junk, markets began to fear for Portugal and Spain.[61]

Finally, on May 2, 2010, the EU and IMF announced a €110 billion package of loans for Greece to be distributed over three years, in exchange for austerity measures intended to cut the deficit by 11% of GDP, on top of a 5% reduction already achieved.[62] Greece also was required to undertake "structural" reforms, enacting more "flexible" labor-relations laws to facilitate hiring and firing workers; lifting regulations deemed inefficient—for example, guaranteeing pharmacies a 35% profit on drugs—selling some state enterprises, and improving the investment climate.[63] Poul Thomsen, I.M.F. mission chief for Greece, declared that the package "will restore confidence."[64]

On May 3, global markets plunged. A week later, the EU and IMF pledged to establish a €750 billion loan package—€440 from European states via the so-called European Financial Stability Facility (EFSF), €60 from the European Commission, and €250 from the IMF—to stem the crisis in Greece and other Eurozone nations.[65] The package was still just a promise. Only a month later did Eurozone finance ministers formally approve the nominal €440 billion EFSF. It took more than a year for the EFSF's approved lending capacity to reach the nominal €440-billion level.[66] And all Eurozone governments would have to approve each individual loan.[67]

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Only the ECB seemed able to move at the speed of markets. The day after the second loan package was announced, the ECB began buying Eurozone sovereign bonds on the secondary market. As the crisis subsided for a while, many saw this action as key.[68] It was also controversial. Axel Weber, the Bundesbank president seen as a likely successor to the ECB presidency, objected in the first of a series of German protests against buying sovereign bonds.[69] He later resigned, it was widely believed, because of opposition to the ECB's sovereign bonds purchases. [70]

Again and again over the next two years, in exchange for loan installments, Greece passed higher taxes, cut budgets deeper, legislated more structural reforms—and confronted fiercer protests. Some EU structural demands were effectively impossible to fulfill. For example, in mid-2011 the EU demanded privatizing €50 billion of state-owned enterprises, including ports, the electric utility, and telecom firms.[71] Even the head of the Athens Chamber of Commerce, a long-time proponent of privatization, said that in the midst of the crisis it would be "really selling the crown jewels at a pittance of their cost."[72] A banker hired to consult on the project pointed out that half the assets were real estate that could not be dumped on the market when purchasers feared Greece might go off the euro and thus their investments might be redenominated in a weaker currency.[73]

EU and IMF projections of Greek growth repeatedly proved wildly overoptimistic. In May 2010, the IMF projected that Greek GDP would fall 2.6% in 2011, then return to growth in 2012.[74] To skip successive revisions, by March 2012, the IMF determined that Greek GDP had fallen 6.8% in 2011 and estimated it would fall another 4.8% in 2012, before—it hoped—returning to growth in 2014.[75] The losses of the Greek economy approached those of the United States during the Great Depression.

As projections of Greek growth sank, projections of debt swelled. The IMF originally expected it would top out at about 150% of GDP and decline to a supposedly sustainable 120% of GDP by 2020.[76] By December 2010, it was to top out at 160% of GDP, by July 2011 at 170% of GDP.[77] By March 2012, had private investors not taken a €100 billion loss, the IMF would have had to project Greek debt as exceeding 200% of GDP—in effect, virtually guaranteed default.[78]

The Crisis Spreads

In July 2010 European bank supervisors declared that nearly all major banks had passed stress tests, but they had not considered sovereign default since they insisted it would never happen.[79] Four months later, in November, Ireland announced that its banks, despite passing the tests in July, had lost about 50% of Irish GDP on bad real-estate loans, which the government would absorb.[80]

Ireland was now forced to follow Greece, requesting a €85 billion loan from the EU and IMF. Irish Prime Minister Brian Cowen pushed a fiscal austerity package through an angry parliament to secure the loan package in December and promised to call parliamentary elections. In February 2011, he lost to Enda Kenny of the Fine Gael Party.[81]

As spring 2011 arrived, Portuguese Prime Minister José Sócrates, hoping to avoid need for an EU-IMF loan, proposed a severe austerity package. After Parliament rejected it and Portuguese banks threatened to stop lending to the government, Sócrates was forced to request the EU-IMF loan.[82] It was approved on May 5 in exchange for the same austerity measures rejected earlier, as well as other reforms such as reducing the public sector's weight in the economy and increasing competition.[83] The Social Democratic Party, led by Pedro Passos Coelho, defeated Sócrates in elections on June 5.

Spain and Italy—too large for the EFSF to handle if they fell into crisis—also faced contagion.[84] In May 2010, the Spanish government barely passed a €15 billion austerity plan to cut the budget deficit

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to 6% of GDP in 2011.[85] Three days later, Spanish debt was downgraded, and the unemployment rate hit 20 percent.[86] By fall the *New York Times* reported that Europeans were "desperate to keep the crisis from overtaking Spain."[87] Periodically abating or surging, market fears drove yields on 10-year Spanish bonds close to 7% in late 2011—about the level at which Greece, Ireland, and Portugal had been forced to seek Troika loans.[88] On November 20, Prime Minister José Luis Rodríguez Zapatero of the Socialist Party was defeated in parliamentary elections by Mariano Rajoy of the Popular Party.

By summer 2011, yields on 10-year Italian bonds touched 6%.[89] Prime Minister Silvio Berlusconi did not help matters by quipping that his austerity-minded finance minister "thinks he's a genius and everyone else is stupid."[90] On August 7, to quell surging market fears, ECB President Jean-Claude Trichet started buying Italian government bonds but demanded in exchange that Berlusconi pass an austerity package and structural economic reforms.[91] For a while yields fell to a less worrisome 5%, but as Berlusconi stalled, yields soared to 7% in November (**Exhibit 14**).

Berlusconi now pressed parliament to pass the austerity measures and, amid the latest round of the corruption scandals that had surrounded his government, was forced to resign.[92] Mario Monti, a former member of the European Commission, became prime minister, leading a cabinet of technocrats.[93] He presented a new austerity package and, in January 2012, more reforms to weaken professional guilds, from notaries to taxi drivers, and rectify the labor laws. Somewhat watered-down but still significant labor reforms finally passed in June.[94]

The European "Firewall"

EU leaders' succession of summit meetings intended to stem the crisis often did little more than create uncertainty, sketch unworkable proposals, or push problems down the road. As Standard & Poor's awarded Greece a CCC credit rating, the lowest in the world, EU leaders prepared for a major July 21, 2011, summit to provide a second Greek loan package and build a "firewall" for Europe.[95] But the summit did little more than set an agenda for debates that would continue well into 2012.

Indeed, after the summit, markets began a precipitous six-month descent (**Exhibit 13**).[96] The Dow Jones Germany stock index, which had held almost even from January 1 until July 22, plummeted 25% during the rest of the year. The Greece index, which by some miracle had only lost about 10% until July 22, sank 50% more by the end of the year. Markets in Italy and France gave up 20% of their value after the summit. The rise in market volatility was even more dramatic. Before July 22, markets were "volatile," gaining or losing more than 2% of their value, on fewer than 1% of trading days. After July 22, markets were volatile on 29% of trading days.[97]

Before building a firewall, EU leaders faced some unfinished business: increasing the EFSF's actual lending capacity to the €440 billion announced in May 2010. They also wanted to give the EFSF new powers to buy sovereign debt on secondary markets, provide credit lines to countries not yet in a rescue program (aka Italy and Spain), and lend to governments to recapitalize banks.[98] National parliaments had to approve these measures. Finally, on October 12, 2011, the Slovakian parliament defeated them—only to approve them two days later after the prime minister resigned.[99]

Now that the €440 billion EFSF was fully established and had its new powers, EU leaders agreed it was inadequate. The European Commission and France wanted to make it a bank. It could then borrow funds from the ECB, which in turn could print unlimited amounts of money.[100] "Implacable German opposition," Reuters reported, nixed this idea.[101]

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Next EU leaders conceived various schemes to leverage the EFSF's €440 billion lending capacity up to €1.25 trillion without actually contributing a cent.[102] One was to create a "special purpose vehicle" allowing China, Russia, or other cash-rich nations to invest in the EFSF.[103] However, it turned out that those cash-rich nations were no more eager than Europe itself to buy peripheral Eurozone sovereign debt. The leveraging plan in the end went nowhere.

Another way to build a firewall might be to sell "eurobonds." Jointly guaranteed by all Eurozone states—every state would have "joint and several" liability for the entire value of each bond—they could provide secure funds to lend threatened governments. The fact that U.S. bonds were guaranteed by a single federal government, not issued piecemeal by one state here and another there, certainly helped the United States borrow at record low interest rates. Italian Finance Minister Giulio Tremonti supported eurobonds.[104] But because German credit would, in fact, largely underpin these bonds, Merkel was "so determined to kill the idea" that she ran over it and then "backed over it repeatedly," Paul Taylor of Reuters quipped. French President Nicolas Sarkozy, eager to stay in her good graces, went along.[105]

EU leaders did pledge to replace the EFSF, set to expire in 2013, with a €500 billion permanent European Stability Mechanism (ESM). The IMF and the Group of 20 leading economic nations wanted Europe to establish an ESM (or combined EFSF and ESM) totaling €1 trillion. If it did, the IMF pledged to add another €500 billion.[106] France and Germany did not want such a large ESM, since they would have to contribute most of the capital. Merkel was said to lack political support for such a sizeable contribution.[107] Some German officials also opposed a larger ESM because they thought the crisis applied useful pressure on spendthrift governments to reform. Finance Minister Wolfgang Schäuble said, "We can only achieve a political union if we have a crisis."[108]

By way of compromise, EU leaders agreed in April 2012 to start funding the ESM in July 2012, so it would overlap the EFSF, set to expire a year later. The headlines said total lending capacity would increase from €500 to €700 billion, but, in fact, €200 would be tied up in existing loans to Greece, Ireland, and Portugal.[109] The ESM was actually inaugurated in October, with the funding of its capital to follow.[110]

Ring-fencing Greece

As a condition for any second official Greek loan, Germany and the Netherlands demanded "private-sector involvement" (PSI), i.e., a loss on private-sector loans.[111] EU leaders' earlier pledge that Greek debt would never be restructured posed a problem: Would anyone believe their continued pledge that no other nations' debt would be restructured? ECB President Trichet fiercely opposed PSI, fearing it would worsen market turmoil.[112] And since the ECB refused to take a loss on Greek bonds—considered an illegal fiscal transfer from other Eurozone nations to Greece—much of the financial industry asked why it should take a hit either.[113]

However, Eurozone legislators saw PSI as necessary to get the expanded EFSF through parliaments; they wanted to see the private sector accept pain, too.[114] As for reversing themselves, EU leaders insisted Greek debt restructuring was "exceptional" and (in case anyone had not heard) "unique."[115] The EU would only provide a second rescue of Greece if debt owners took a hit.[116]

PSI raised concern over some €50 billion of credit-default swaps (CDS)—insurance bondholders had taken out against Greek default.[117] If a weak financial firm had insured more than it could afford, as the American International Group (AIG) fatefully had before the 2008 U.S. crisis, it might trigger panic. To avert this eventuality, EU leaders proposed that the PSI be "voluntary." If banks were

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pressured to take "voluntary" losses, the CDS might not be triggered. Or, then again, they might. "There's not any clarity here because people don't know," said Christopher Whalen, editor of *Institutional Risk Analyst*.[118] Although large banks generally agreed to participate in a "voluntary" PSI, many hedge funds and other non-bank holders of Greek debt threatened not to.[119]

Meanwhile, as the Greek economy kept doing worse than projected and Greek debt kept soaring higher than projected, Merkel and Sarkozy said Greece would not receive the fall 2011 installment of the existing loan package, much get less a second package, unless it accepted more austerity.[120] Greece would cut pensions 20%, put 30,000 government workers on 60 percent pay for a year and then dismiss them, and impose large real-estate surtaxes that took no account of income.[121]

Faced with fierce political protest, on October 31, Prime Minister George Papandreou suggested a referendum to let the "wise and capable" Greek people decide about the austerity measures—and, fundamentally, on staying in the euro.[122] Floyd Norris of the *New York Times* wondered why not give democracy a try.[123]

The idea of letting the Greek people decide prompted Merkel and Sarkozy to speak the unspeakable: Greece could be booted out of the euro.[124] When even the Greek finance minister lambasted the referendum, Papandreou agreed to resign.[125] A "unity" government was cobbled together to see Greece through passing the austerity packages and second loan. Lucas Papademos, a former ECB vice president and Harvard visiting professor, was named interim prime minister, overseeing a cabinet of feuding politicians.[126] Elections would be held in the spring.

Amid continual worsening of the Greek economy and debt, PSI became a moving target. Private debt holders were told in July 2011 that they must take a 21% loss, in October that they must take a 50% loss, and, finally, in March 2012 that they must take a 74% loss.[127] On March 9, 2012, investors holding more than 85% of Greek debt "voluntarily" accepted these losses, Greece forced owners of other bonds governed by Greek law to participate, the CDS were triggered, and no catastrophe ensued.[128] In fact, triggering the CDS reassured owners of CDS on other sovereign debt.[129] Athens' €350 billion of debt was reduced by €100 billion.[130] Old Greek bonds were swapped for new bonds that were worth far less but governed by English law, so bondholders enjoyed greater legal security.[131] The ECB refused to take a loss but agreed to transfer any profits it might make on Greek government bonds to the Greek government.[132]

Yet again the EU demanded more austerity and structural reforms, ranging from overhauling the judicial system to buying new computers for tax collectors.[133] Of singular importance was cutting the minimum wage, on which all private-sector wages were based, by 22%, on top of an estimated 15% decline since 2009.[134] In effect, the wage demand simulated what would have happened to a country with its own currency. As had sometimes occurred in Latin America, the currency would crash, lowering real wages and, in time, making exports more competitive. Since Greece was on the euro, the currency could not crash, and even deep unemployment had yielded modest wage declines. This time, Europe made sure Greek wages fell.

Crisis Threatens the Banks

The European banking system, particularly in peripheral nations, faced deep trouble. After a second round of stress tests on July 15, 2011, the European Banking Authority declared that only 8 of 90 banks were too weak to survive shocks, and would require a mere €2.5 billion to shore up their Tier 1 capital to 6% of assets.[135] Analysts were doubtful. IMF Managing Director Christine Lagarde hinted that European banks might need a €200 billion, rather than a €2.5 billion, recapitalization.

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Having become highly leveraged during the pre-crisis years, banks were facing a vicious cycle of deleveraging. In the initial step, sovereign bonds lost value, eroding banks' capital. To rebuild capital, which includes cash, they were forced to sell assets, in particular sovereign bonds of foreign governments. (Their fates were so closely tied to fates of their own governments that they hesitated to sell their own governments' bonds.) The result was that, across the economy, sovereign bond prices fell further. Thus, falling bond prices prompted bond sales, and bond sales caused further price falls. Bank capital eroded, and governments had to pay higher interest rates to roll over debt.

A related vicious cycle involved the private sector. For one thing, as government-bond yields rose, interest rates on business loans rose in parallel, squeezing firms and eroding growth. For another thing, banks tried to reduce their assets in part by curtailing loans to businesses. The effect was to weaken the economy, erode tax receipts, worsen the sovereign-debt crisis, and put additional pressure on banks. Everything was working in the wrong direction.

On the other side of the ledger book, banks also had trouble retaining funds. Nations in crisis are doubtful places to park money, so foreigners became reluctant to deposit money in peripheral banks or buy their bonds. And existing deposit holders made withdrawals, moving money to seemingly more stable banks in Germany or elsewhere. In 2011, some €32 billion, or 16% of deposits, were withdrawn from Greek banks.[136]

In July, 2011, EU officials rejected Lagarde's suggestion that banks needed to raise capital to buffer against losses.[137] But by September, they reversed themselves, agreeing that banks must raise substantial capital.[138] As often, they could not agree on means. Sarkozy did not want to have to inject French government funds into French banks for fear of a credit downgrade for France, but Merkel did not want to use EU funds to bolster the foreign competitors of German banks.[139] The compromise was that banks must raise their own capital. If they could not, their governments must help them. If governments could not, they might request loans from the EFSF. On October 26, EU heads of state required banks to raise Tier 1 capital by €100 billion euros by mid-2012.[140]

A New Fiscal Pact

Despite their previous summits and declarations, EU leaders faced a precarious situation in late 2011. As mentioned, yields on 10-year Spanish bonds approached 7% and those on Italian bonds surpassed 7% in November, levels at which other nations had been forced to seek Troika loans. But the EFSF did not have the resources to stave off a Spanish or Italian default. And Standard & Poor's warned it might downgrade 15 Eurozone countries in one fell swoop if EU leaders did not deliver a convincing plan to halt the crisis. [141]

Thus, EU leaders met on December 9 to deliver what they hoped would be a convincing plan: a new fiscal treaty to rigorously enforce the much-violated Growth and Stability Pact. In addition to meeting existing requirements—deficits no higher than 3% of GDP and debt no higher than 60% of GDP—governments would now be required to limit "structural" deficits. The structural deficit is meant to gauge the underlying deficit, discounting business cycle fluctuations. For example, if an economy is in recession, welfare payments naturally rise and tax receipts fall. To discount these effects, economists try to estimate what payments and taxes would be—hence what the "structural" deficit would be—if the economy were at a long-run equilibrium. EU announced that nations must adopt a rule to keep the structural deficit under 0.5% of GDP, and must legally incorporate an automatic correction mechanism "at constitutional level or equivalent" if it were violated.

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The treaty raised a host of complications. British prime minister David Cameron rejected it since it also included a tax on financial transactions that he deemed unfair to the U.K. financial industry.[142] Britain's absence left doubts as to what legal status other EU nations had to ratify and enforce the agreement.[143] François Hollande, the Socialist Party candidate challenging Sarkozy in spring 2012 French presidential elections, said he would demand changes in the treaty, whereupon Sarkozy declared he would put it to a referendum.[144] Even Merkel needed the opposition Social Democrats to secure a two-thirds vote for the measure in the parliament.[145] Some EU officials worried that, as ratifying the treaty encountered obstacles, the process would only add to political uncertainty and undermine confidence.[146]

The content of the agreement posed complications, too. *Financial Times* columnist Martin Wolf pointed out that the "structural" deficit is a theoretical concept. It can be pretty well calculated in normal times, but real-time calculation is difficult and least reliable during a crisis. For example, in October 2007, the IMF calculated that the Irish structural budget was balanced, but later said it had been running an 8% of GDP structural deficit.[147] Wolf asked how the rule would be enforced: "On deeply uncertain estimates of structural deficits, the Commission—a body of unelected bureaucrats—would impose sanctions on elected governments, when the latter are under great pressure. What is the Commission going to do if they still fail to comply?"[148]

As well, numerous economists and officials, such as IMF Managing Director Lagarde and U.S. Treasury Secretary Geithner, repeatedly warned that to clamp down on deficits precisely in the midst of a crisis could deepen the recession, and worsen budget deficits.[149] German Finance Minister Wolfgang Schäuble made the counterargument: "The main reason for the lack of demand is the lack of confidence; the main reason for the lack of confidence is the deficits and public debts which are seen as unsustainable."[150] Some economists even proposed a theory that if fiscal austerity restored confidence, private spending would counterbalance the drop in public spending, the economy would grow, and deficits would fall.[151]

On March 2, 2012, all EU members except Britain and the Czech Republic signed the treaty, though ratification would be months off, at best. Spanish Prime Minister Mariano Rajoy signed the treaty and, incidentally, announced that Spain's deficit would not be 4.4% of GDP, as pledged, but rather 5.8% of GDP, because growth had proved worse than expected.[152] Though in theory the treaty stripped sovereigns of latitude in setting deficits, Rajoy added that the deficit decision was a matter of national sovereignty.[153] Eurozone finance ministers later bargained him down to 5.3% of GDP.

The "Wall of Money"

On December 8, 2011, as EU leaders were gathering for the summit to try to cement their new fiscal pact, ECB President Mario Draghi formally proposed a so-called "long term refinancing operation" (LTRO) to the board.[154] The ECB would offer Eurozone banks a "wall of money": as large a volume of three-year, low interest-rate loans as they wanted. He thought the banks would use the money to lend to their governments, bringing down yields. They would still make a good profit on the spread between the higher interest rates they would receive and the lower rates the ECB would charge. And they had every reason to support their governments; if the governments defaulted, they would go the same way. Draghi also hoped banks would use ECB resources to lend to private firms and strengthen the economy.

This plan was, moreover, politically feasible. Conceding that the ECB was responsible for providing liquidity to banks, German members did not protest, and the board approved. On December 21 more than 500 banks took out €489 billion in long-term loans from the ECB. They used

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much of the money to replace short-term loans, but about €190 billion constituted fresh borrowing.[155] The ECB provided a second wall of money on February 29, as banks borrowed an additional €529.5 billion, about €310 billion of it constituting net new liquidity.[156] Two thirds of the liquidity was taken up by Italian and Spanish Banks. Total ECB lending reached more than €3 trillion euros—higher than U.S. Federal Reserve lending ever had.[157] (See **Exhibit 15**.)

The banks did lend to sovereigns, as hoped. As Italy and Spain carried off successful auctions, their bond yields fell back into the 5% range.[158] But some doubted the longer-term effects. European banks' lending to their sovereigns might just be a way to keep sovereigns afloat even as the banks piled on toxic debt and hoped for salvation.[159] "Zombie banks support zombie companies" and, by analogy, zombie governments, said Nicolas Véron of Bruegel.[160] Moreover, European bank lending to firms and consumers was not picking up (See **Exhibit 16**) and there were material differences in the cost of loans across Europe, with small to medium size firms in Spain and Italy paying up to 2 percentage points (200 basis points) more than their French or German competitors.

By spring Europe clearly was not growing. IMF Director Lagarde warned that "across-the-board, across-the-continent, budgetary cuts will only add to recessionary pressures"—and, as a result, market concern about sovereign debts.[161] Prime Minister Rajoy announced that Spain would run a larger deficit than projected because growth was slower than projected.[162] Portugal, considered a model of structural reform, seemed headed deeper into debt as its economy weakened.[163] The IMF warned that Italian GDP would decline 1.9% in 2012.[164]

In June 2012, as the boost from the wall of money dissipated, and euro optimism sank along with it, European leaders announced what was hailed as a "game changer."[165] The permanent sovereign rescue fund, the ESM, would have power to inject capital directly into banks, breaking the vicious circle in which sovereigns took on debt to rescue banks, and banks weakened as they lent to sovereigns. However, Germany won a concession stipulating that a Eurozone bank regulator must be established before the ESM could recapitalize banks.

In December European leaders agreed to establish that regulator. The ECB would monitor reserves and lending of banks with assets of more than €30 billion or more than 20% of national GDP and could require changes in unsound practices.[166] For the most part, national regulators would still handle smaller banks. However, EU officials themselves warned that supervision would remain ineffective unless some future agreement gave the ECB authority to stop a crisis by shutting down or recapitalizing weak banks.[167] Moreover, it was far from clear that the ESM would be used to sever the perilous tie between governments and national banks. Echoing German concerns, the European Commission proposed that no ESM funds be given to banks that had less capital than the regulatory minimum—a definition that could apply to many troubled banks.[168]

Draghi's bold gambit in July to fortify the euro by pledging that the ECB would do "whatever it takes" did, indeed, mean giving the bank authority to buy large quantities of sovereign bonds in what came to be called its "outright monetary transactions" (OMT) program. Next he had to get his proposal through the ECB board, which reportedly had not been warned.[169] Jens Weidmann, president of the German Bundesbank, flatly opposed the idea, but German Chancellor Merkel backed it, with a condition: indebted nations must undertake the austerity and structural reforms she had pushed all along before the ECB could buy their bonds.[170] In September, Draghi announced that the ECB would buy potentially unlimited amounts of sovereign debt of up to three-year maturities to prevent "distortions in financial markets"—unjustifiably high interest rates—but only after governments entered standard programs to implement fiscal austerity and structural reforms.[171] Neither Spain nor Italy showed interest in entering such a program, at least through late 2012.[172]

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By fall 2012, the EU and IMF agreed that Greece needed a second round of debt relief to avoid default but disagreed about how much relief.[173] The IMF wanted a higher level of relief, which would probably force writedowns on official loans, while EU leaders balked at such politically unpopular writedowns. On November 27 a compromise was reached, including reducing interest rates to only 0.5 percentage points (50 basis points) above the interbank rate, lengthening maturities by 10 to 15 years, returning ECB profits on its Greek loans to Greece, and other measures. Greek debt was projected to fall to 124% of GDP in 2020 and "substantially below" 110% of GDP in 2022—a goal seen as likely to require more debt relief in the future.

What Next?

In spring 2013, fears that the euro would break up diminished, and European markets were up from the depths of 2012 (**Exhibit 13**), yet the whole southern Eurozone had settled into prolonged recession, while the core stood at risk of recession.[174] Italy's GDP had fallen 7 percent since 2007 and industrial production by a quarter, while unemployment had doubled.[175] In March the Troika demanded what "almost borders on ritual," as a *New York Times* reporter put it: more Greek austerity in exchange for a new loan tranche.[176] Faced with banking failures, Cyprus sought a €10 billion Eurozone loan. In a hardly unfamiliar pattern, on-again, off-again negotiations continued until the night before the whole situation would have collapsed. This time, private bondholders and depositors with more than €100,000 in their accounts took material losses, and Cyprus got its loan.

In Italy's February parliamentary elections, Beppe Grillo, a comedian waging an anti-euro and anti-system campaign, won 54 seats, far ahead of Mario Monti's dismal 19 seats.[177] The center-left Democrats won a narrow plurality. Berlusconi rose from disgrace, demanding tax cuts and lambasting German "hegemony" in enforcing austerity. His People of Liberty party came in a close second to the Democrats.[178] After the election, the two major parties refused to form a coalition, and Grillo refused to form a coalition with either. Monti struggled as interim prime minister without support. The only clear thing was Italians' fury at years of austerity.

In the face of political impasse over a successor, the 87-year-old President Napolitano was finally prevailed upon to accept a second term. Nearly in tears in his acceptance speech, he excoriated politicians for their "fatal deadlock" and demanded that a coalition be formed within days or implicitly threatened to resign.[179] The Democrats and the People of Liberty did form a coalition, headed by Democrats leader Enrico Letta. How long the coalition would hold was anyone's guess.

Political compromise among European leaders, as well as within nations, seemed essential if the Eurozone was ever to emerge from crisis. In justifying one of its many downgrades of Eurozone debt, Standard & Poor's had cited the "open and prolonged dispute among European policymakers."[180] There were, indeed, signs that European Commission President Barroso and German Chancellor Merkel (despite adamant Bundesbank opposition) saw a need to let up—modestly—on austerity.

The project of European integration had faced crises before, and each time its response had ultimately strengthened the union. Could the European Union strengthen itself this time?

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Exhibit 1a Fiscal balance, percent of GDP, Eurozone and selected other nations, Maastricht violations highlighted

	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Euro 17 average	-1.5	-0.1	-1.9	-2.6	-3.1	-2.9	-2.5	-1.3	-0.7	-2.1	-6.4	-6.2	-4.2	-3.7
Germany	-1.6	1.1	**-3.1**	**-3.8**	**-4.2**	**-3.8**	**-3.3**	-1.6	0.2	-0.1	**-3.1**	**-4.1**	-0.8	0.2
France	-1.8	-1.5	-1.5	-3.1	**-4.1**	**-3.6**	-2.9	-2.3	-2.7	**-3.3**	**-7.5**	**-7.1**	**-5.3**	**-4.8**
Greece			**-4.5**	**-4.8**	**-5.6**	**-7.5**	**-5.2**	**-5.7**	**-6.5**	**-9.8**	**-15.6**	**-10.7**	**-9.5**	**-10.0**
Ireland	2.6	**4.7**	0.9	-0.4	0.4	1.4	1.7	2.9	0.1	**-7.4**	**-13.9**	**-30.8**	**-13.4**	**-7.6**
Portugal	-3.1	-3.3	**-4.8**	-3.4	-3.7	**-4.0**	**-6.5**	**-4.6**	**-3.1**	**-3.6**	**-10.2**	**-9.8**	**-4.4**	**-6.4**
Spain	-1.2	-0.9	-0.5	-0.2	-0.3	-0.1	1.3	2.4	1.9	**-4.5**	**-11.2**	**-9.7**	**-9.4**	**-10.6**
Italy	-1.9	-0.8	**-3.1**	**-3.1**	**-3.6**	**-3.5**	**-4.4**	**-3.4**	-1.6	-2.7	**-5.5**	**-4.5**	**-3.8**	**-3.0**
Austria	-2.3	-1.7	0.0	-0.7	-1.5	**-4.4**	-1.7	-1.5	-0.9	-0.9	**-4.1**	**-4.5**	-2.5	-2.5
Belgium	-0.6	0.0	0.4	-0.1	-0.1	-0.1	-2.5	0.4	-0.1	-1.0	**-5.6**	**-3.8**	**-3.7**	**-3.9**
Finland	1.7	7.0	5.1	4.2	2.6	2.5	2.9	4.2	5.3	4.4	-2.5	-2.5	-0.8	-1.9
Luxembourg	3.4	6.0	6.1	2.1	0.5	-1.1	0.0	1.4	3.7	3.2	-0.8	-0.9	-0.2	-0.8
Netherlands	0.4	2.0	-0.2	-2.1	**-3.1**	-1.7	-0.3	0.5	0.2	0.5	**-5.6**	**-5.1**	**-4.5**	**-4.1**
Slovenia									0.0	-1.9	**-6.2**	**-5.9**	**-6.4**	**-4.0**
Cyprus										0.9	**-6.1**	**-5.3**	**-6.3**	**-6.3**
Malta										**-4.6**	**-3.7**	**-3.6**	**-2.8**	**-3.3**
Slovakia											**-8.0**	**-7.7**	**-5.1**	**-4.3**
Estonia													1.2	-0.3
Britain	0.9	3.6	0.5	-2.1	-3.4	-3.5	-3.4	-2.7	-2.8	-5.1	-11.5	-10.2	-7.8	-6.3
Japan	-7.1	-7.4	-6.0	-7.7	-7.7	-5.9	-4.8	-1.3	-2.1	-1.9	-8.8	-8.4	-9.3	-9.9
United States	0.7	1.5	-0.6	-4.0	-5.0	-4.4	-3.3	-2.2	-2.9	-6.6	-11.9	-11.4	-10.2	-8.5

Exhibit 1b Primary balance, percent of GDP, selected Eurozone nations

	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Euro 17 average	2.5	3.8	1.8	0.8	0.1	0.2	0.5	1.5	2.3	0.9	-3.5	-3.4	-1.1	-0.6
Germany	1.6	4.3	0.0	-0.9	-1.1	-0.9	-0.5	1.2	3.0	2.7	-0.4	-1.6	1.8	2.6
France	1.2	1.4	1.5	-0.2	-1.3	-0.9	-0.3	0.3	0.0	-0.4	-5.1	-4.7	-2.7	-2.3
Greece	N.A.	3.6	2.0	0.7	-0.7	-2.6	-0.7	-1.4	-2.0	-4.8	-10.5	-4.9	-2.4	-5.0
Ireland	4.9	6.7	2.4	1.1	1.7	2.5	2.7	4.0	1.1	-6.0	-11.8	-27.7	-10.0	-3.9
Portugal	-0.2	-0.3	-1.8	-0.6	-1.0	-1.4	-4.0	-1.8	-0.2	-0.6	-7.3	-7.0	-0.4	-2.0
Spain	2.3	2.3	2.5	2.5	2.0	1.9	3.1	4.0	3.5	-2.9	-9.4	-7.7	-7.0	-7.7
Italy	4.6	5.4	3.1	2.5	1.5	1.2	0.2	1.2	3.4	2.5	-0.8	0.1	1.2	2.5

Note: Data for Eurozone nations are provided only beginning with their date of entry.

Sources: EU nations (including UK): Eurostat, updated Apr. 26, 2013, General government, "Net lending (+)/Net borrowing (-) under the EDP (Excessive Deficit Procedure)" and "Primary balance." U.S. and Japan: OECD Economic Outlook, Volume 2012 Issue 2, no. 92, Statistical Annex, version 1, updated Dec. 4, 2012, Annex Table 27, "General government financial balances."

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Exhibit 2 Government debt, percent of GDP, Eurozone and selected other nations, Maastricht violations highlighted

	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Euro 17	71.7	69.2	68.2	68.0	69.2	69.6	70.3	68.6	66.4	70.2	80.0	85.4	87.3	90.6
Germany	**61.3**	**60.2**	59.1	**60.7**	**64.4**	**66.2**	**68.5**	**68.0**	**65.2**	**66.8**	**74.5**	**82.4**	**80.4**	**81.9**
France	58.9	57.3	56.9	58.8	**62.9**	**64.9**	**66.4**	**63.7**	**64.2**	**68.2**	**79.2**	**82.4**	**85.8**	**90.2**
Greece			**103.7**	**101.7**	**97.4**	**98.6**	**100**	**106.1**	**107.4**	**112.9**	**129.7**	**148.3**	**170.3**	**156.9**
Ireland	47.0	35.1	35.2	32.0	30.7	29.5	27.3	24.6	25.1	44.5	**64.8**	**92.1**	**106.4**	**117.6**
Portugal	51.4	50.7	53.8	56.8	59.4	61.9	**67.7**	**69.4**	**68.4**	**71.7**	**83.7**	**94**	**108.3**	**123.6**
Spain	**62.4**	59.4	55.6	52.6	48.8	46.3	43.2	39.7	36.3	40.2	53.9	**61.5**	**69.3**	**84.2**
Italy	**113.1**	**108.6**	**108.3**	**105.4**	**104.1**	**103.7**	**105.7**	**106.3**	**103.3**	**106.1**	**116.4**	**119.3**	**120.8**	**127**
Austria	**66.8**	**66.2**	**66.8**	**66.2**	**65.3**	**64.7**	**64.2**	**62.3**	**60.2**	**63.8**	**69.2**	**72.0**	**72.5**	**73.4**
Belgium	**113.6**	**107.8**	**106.5**	**103.4**	**98.4**	**94.0**	**92.0**	**88.0**	**84.0**	**89.2**	**95.7**	**95.5**	**97.8**	**99.6**
Finland	45.7	43.8	42.5	41.5	44.5	44.4	41.7	39.6	35.2	33.9	43.5	48.6	49.0	53
Luxembourg	6.4	6.2	6.3	6.3	6.1	6.3	6.1	6.7	6.7	14.4	15.3	19.2	18.3	20.8
Netherlands	**61.1**	53.8	50.7	50.5	52	52.4	51.8	47.4	45.3	58.5	**60.8**	**63.1**	**65.5**	**71.2**
Slovenia									23.1	22.0	35.0	38.6	46.9	54.1
Cyprus										48.9	58.5	**61.3**	**71.1**	**85.8**
Malta										**60.9**	**66.4**	**67.4**	**70.3**	**72.1**
Slovakia											35.6	41.0	43.3	52.1
Estonia													6.2	10.1
Britain	43.7	41.0	37.7	37.7	39.1	41.0	42.2	43.3	44.2	52.7	67.8	79.4	85.5	90.0
Japan	129.0	137.6	144.7	153.5	158.3	166.3	169.5	166.8	162.4	171.1	188.7	192.7	205.3	214.3
United States	60.5	54.5	54.4	56.8	60.2	67.8	67.4	66.1	66.5	75.4	88.8	97.8	102.2	109.8

Notes:
Data for Eurozone nations are provided only beginning with their date of entry.
Gross debt data for the United States and Japan are not strictly comparable to EU data; the procedures determining what counts as debt differ.

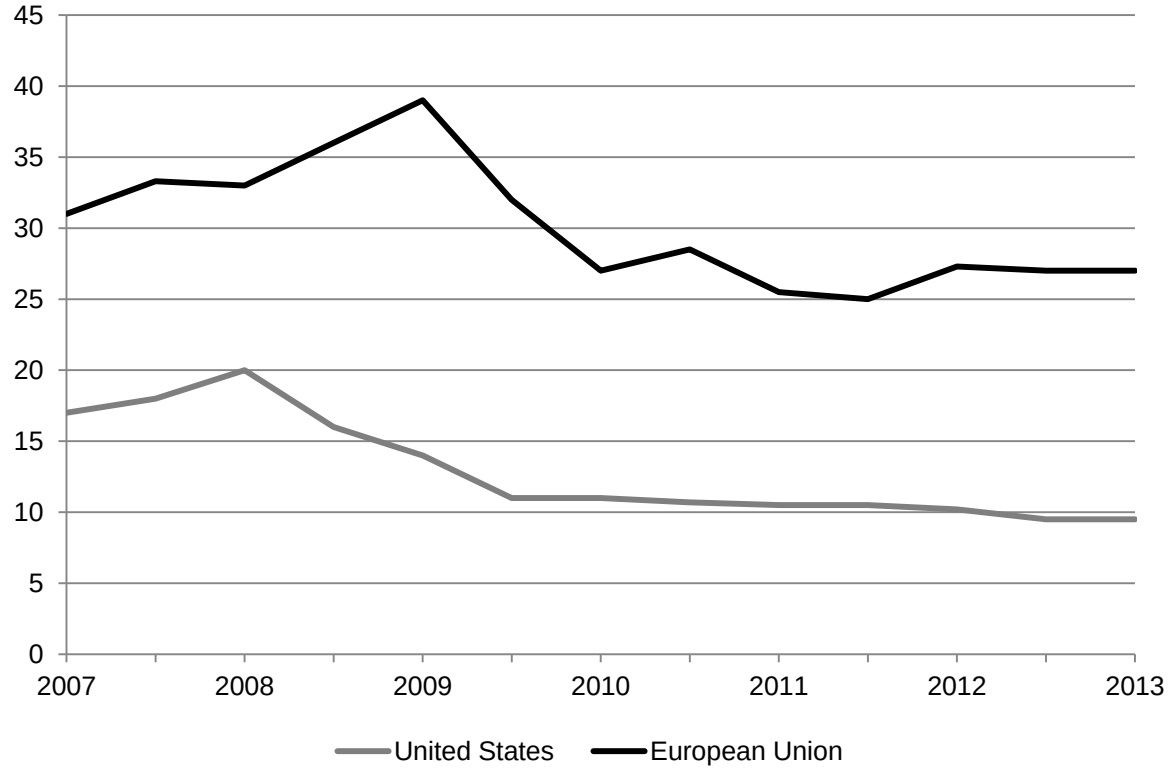
Sources: EU nations (including UK): Eurostat, updated Apr. 26, 2013, General government, "Government consolidated gross debt." U.S. and Japan: OECD Economic Outlook, Volume 2012 Issue 2, no. 92, Statistical Annex, version 1, updated Dec. 4, 2012, Annex Table 32, "General government gross financial liabilities."

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Exhibit 3 Bank leverage: assets divided by equity, sample of major national banks



Source: Created by authors from Adrian van Rixtel and Gabriele Gasperini, "Financial crises and bank funding: recent experience in the euro area," Bank for International Settlements Working Paper 406, 2013, "Balance sheet leverage ratios," p. 24.

Exhibit 4 Which sectors hold government debt, percent of total debt, mid-2011

	Domestic banks	Other public institutions	ECB and national central bank	Other residents of country	Non-residents
Germany	22.9	0	0.3	14.1	62.7
France	14.0	-	n/a	29.0	57.0
Greece	19.4	10.1	25.5	6.5	38.5
Ireland	16.9	0.9	16.1	2.43	63.8
Portugal	22.4	-	12.0	13.5	52.1
Spain	28.3	-	8.3	30.2	33.2
Italy	27.3	-	9.3	26.7	36.7
United Kingdom	10.7	0.1	19.4	39.5	30.2
United States	2.0	35.5	11.3	19.9	31.4

Note: In the case of the United States, "other public institutions" include principally Social Security, as well as some other government agencies.

Source: Jean Pisani-Ferry, "The Euro Crisis and the New Impossible Trinity," Bruegel Policy Contribution, issue 2012/01, January 2012. Bruegel.

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Exhibit 5 Comparative labor-market characteristics

	Employment protection legislation, index from 0 (weakest) to 6 (strongest)		Coverage of collective bargaining agreements, percent of workforce	Unemployment insurance and other assistance to unemployed as percent of prior earnings		Marginal tax "wedge" on labor (percent of additional compensation that is taxed)
	Protection for regular (non-temporary) employees	Additional protection for collective dismissals		Short-term	Long-term	
Germany	3.0	3.8	63	67	59	63
France	2.5	2.1	90	72	54	52
Greece	2.3	3.3	65	54	4	51
Ireland	1.6	2.4	44	55	75	55
Portugal	3.6	1.9	65	79	43	47
Spain	2.5	3.1	88	72	37	48
Italy	1.8	4.9	80	70	N.A.	54
Sweden	2.9	3.8	91	67	63	48
United States	0.2	2.9	14	55	25	34

Notes: The OECD assigns Italy zero long-term employment benefits; it is not clear why so the authors inserted N.A. The marginal tax "wedge" on labor is for an individual without children with average earnings (no family data available). All data are for 2008 except marginal tax wedge data are for 2009.

Source: Adapted from OECD, "Economic Policy Reforms 2011: Going for Growth," 2011, chapter 3, updated Mar. 9, 2011, Figures 3.2, 3.4, 3.8, and 3.9.

Exhibit 6 Business regulation, index from 0 (least restrictive) to 6 (most), except as noted

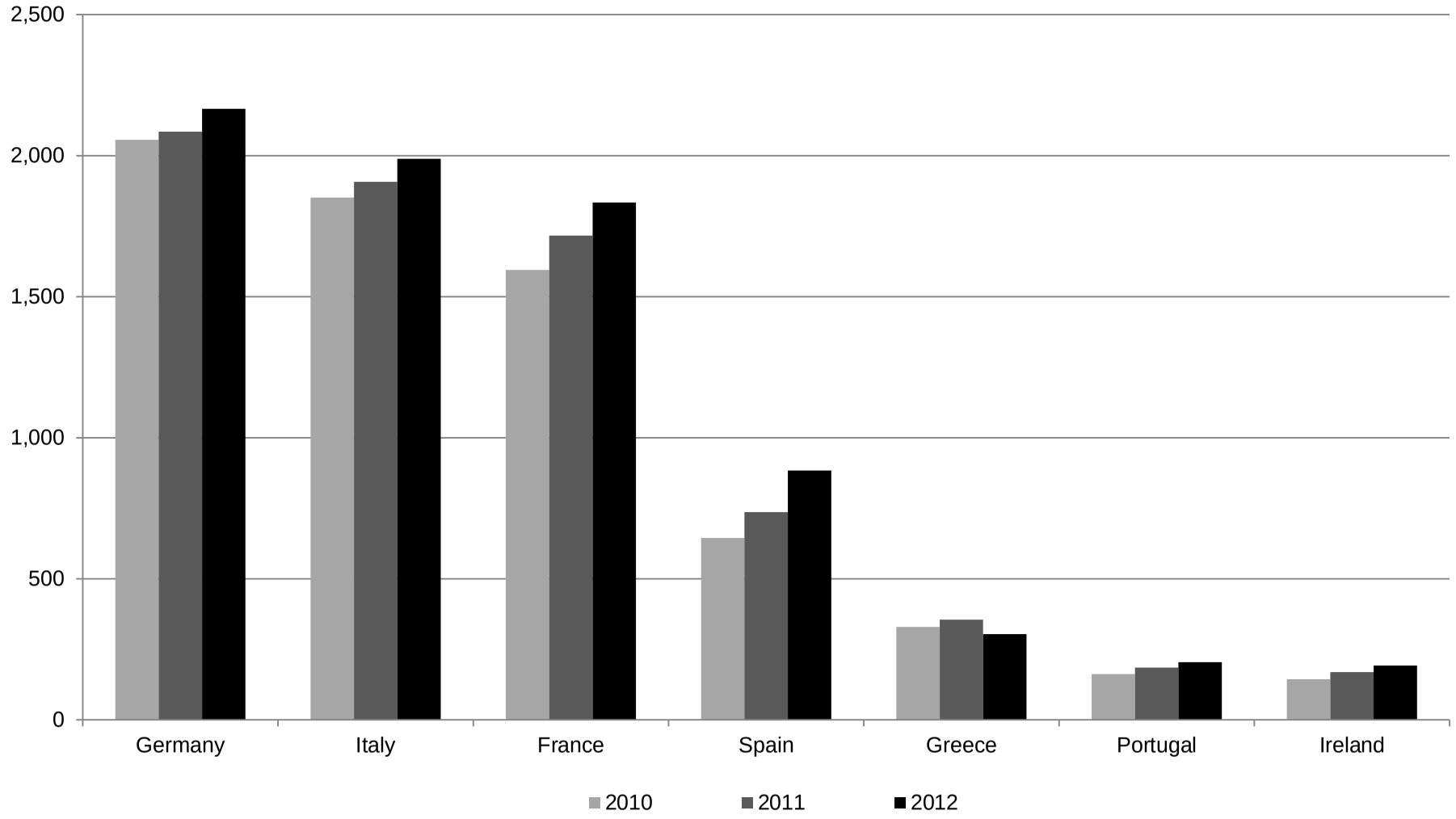
	Restrictiveness of product market regulation	Regulation of retail sector	Regulation of professional services	State involvement in business	Average administrative burdens on corporations and start-ups	Ease of doing business rank out of 183 nations, in 2011
Germany	1.3	2.4	2.9	1.2	0.5	19
France	1.5	3.1	2.1	1.6	1.4	26
Greece	2.4	3.5	2.8	3.7	2.8	101
Ireland	0.9	1.0	0.9	0.4	0.5	8
Portugal	1.4	3.0	2.5	1.6	1.9	30
Spain	1.0	2.7	2.1	1.0	2.4	45
Italy	1.4	2.6	3.2	1.3	1.8	83
Sweden	1.3	0.5	0.6	0.7	1.0	9
United States	0.8	2.6	1.1	0.9	1.0	4

Note: All data except ease of doing business ranking are for 2008.

Sources: OECD, "Going for Growth, 2011," chapter 3, Figures 3.10, 3.11, 3.12, and 3.19. Ease of doing business data from World Bank, "Doing Business 2012," Table 1.1.

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Exhibit 7 Government debt 2010-2012, billions of euros



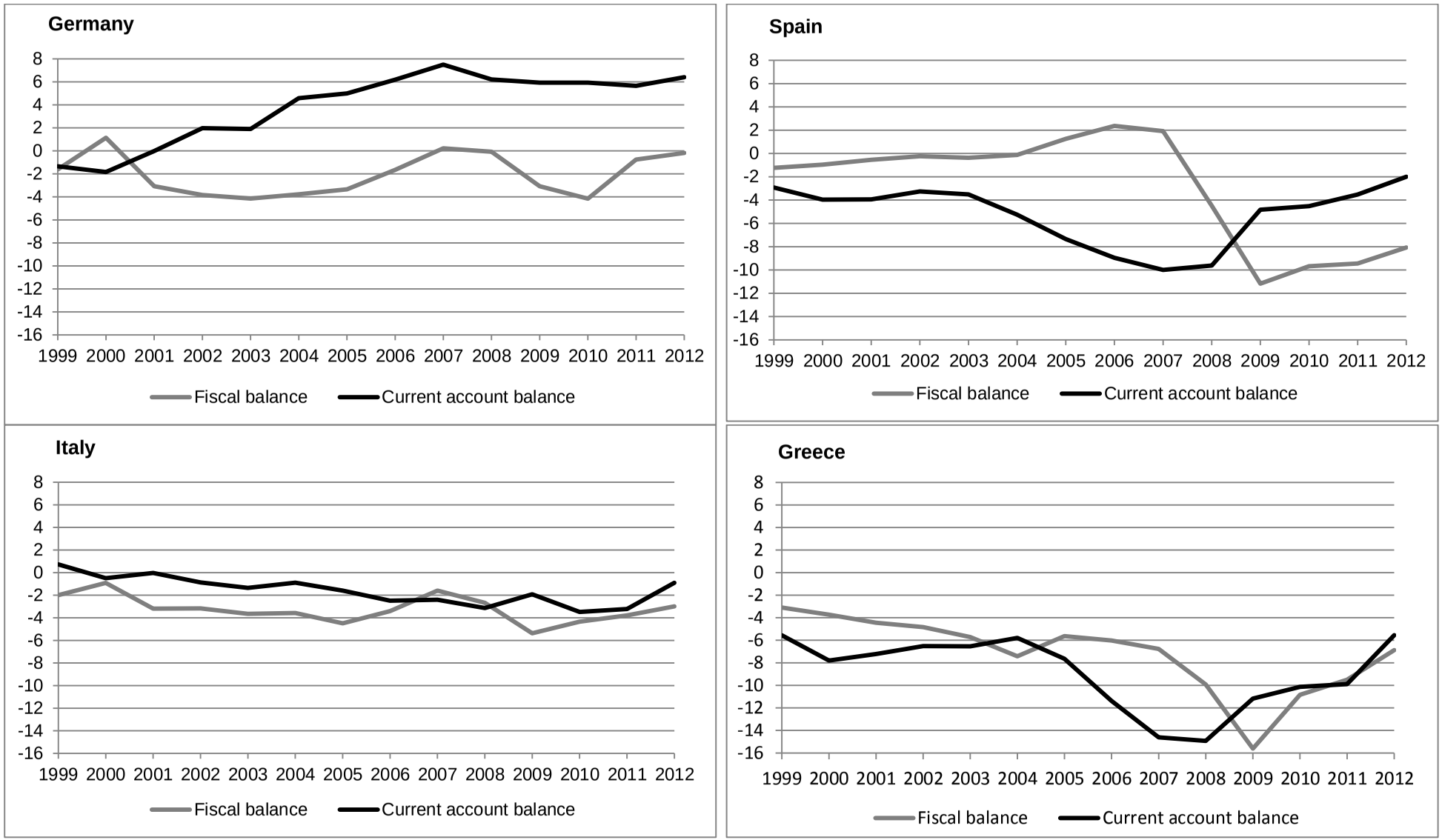
Source: Eurostat, "Government consolidated gross debt," General government, updated April 26, 2013.

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Exhibit 8 Fiscal and current account balances, selected Eurozone nations, percent of GDP



Source: Adapted from OECD Economic Outlook, volume 2012, issue 2, no. 92, Statistical Annex, version 1, updated Dec 4, 2012, Annex Table 27, "General government financial balances," and Annex Table 51, "Current account balances as a percentage of GDP."

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Exhibit 9 Real GDP growth, selected Eurozone countries

	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Euro 17	2.9	3.8	2	0.9	0.7	2.2	1.7	3.2	3	0.4	-4.4	2	1.4	-0.6
Germany	1.9	3.1	1.5	0	-0.4	1.2	0.7	3.7	3.3	1.1	-5.1	4.2	3	0.7
France	3.3	3.7	1.8	0.9	0.9	2.5	1.8	2.5	2.3	-0.1	-3.1	1.7	1.7	0
Greece	3.4	3.5	4.2	3.4	5.9	4.4	2.3	5.5	3.5	-0.2	-3.1	-4.9	-7.1	-6.4
Ireland	11.1	10.7	5.3	5.6	3.9	4.4	5.9	5.4	5.4	-2.1	-5.5	-0.8	1.4	0.9
Portugal	4.1	3.9	2	0.8	-0.9	1.6	0.8	1.4	2.4	0	-2.9	1.9	-1.6	-3.2
Spain	4.7	5	3.7	2.7	3.1	3.3	3.6	4.1	3.5	0.9	-3.7	-0.3	0.4	-1.4
Italy	1.5	3.7	1.9	0.5	0	1.7	0.9	2.2	1.7	-1.2	-5.5	1.7	0.4	-2.4

Note: Greek and Portugese growth data are provisional for some years; French growth is projected for 2012.

Source: Eurostat, "GDP and main components," volumes [nama_gdp_k], Gross domestic product at market prices, Millions of national currency, chain-linked volumes, reference year 2005, percentage change on previous period Updated Apr. 29, 2013, extracted Apr. 29, 2013.

Exhibit 10 Unemployment rate, selected Eurozone countries

	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
Euro 17	9.6	8.7	8.1	8.5	9	9.3	9.2	8.5	7.6	7.6	9.6	10.1	10.2	11.4
Germany	8.6	8	7.9	8.7	9.8	10.5	11.3	10.3	8.7	7.5	7.8	7.1	5.9	5.5
France	10.4	9	8.2	8.3	8.9	9.3	9.3	9.2	8.4	7.8	9.5	9.7	9.6	10.2
Greece	12	11.2	10.7	10.3	9.7	10.5	9.9	8.9	8.3	7.7	9.5	12.6	17.7	24.3
Ireland	5.6	4.2	3.9	4.5	4.6	4.5	4.4	4.5	4.7	6.4	12	13.9	14.7	14.7
Portugal	5	4.5	4.6	5.7	7.1	7.5	8.6	8.6	8.9	8.5	10.6	12	12.9	15.9
Spain	13.2	11.7	10.5	11.4	11.4	10.9	9.2	8.5	8.3	11.3	18	20.1	21.7	25
Italy	10.9	10	9	8.5	8.4	8	7.7	6.8	6.1	6.7	7.8	8.4	8.4	10.7

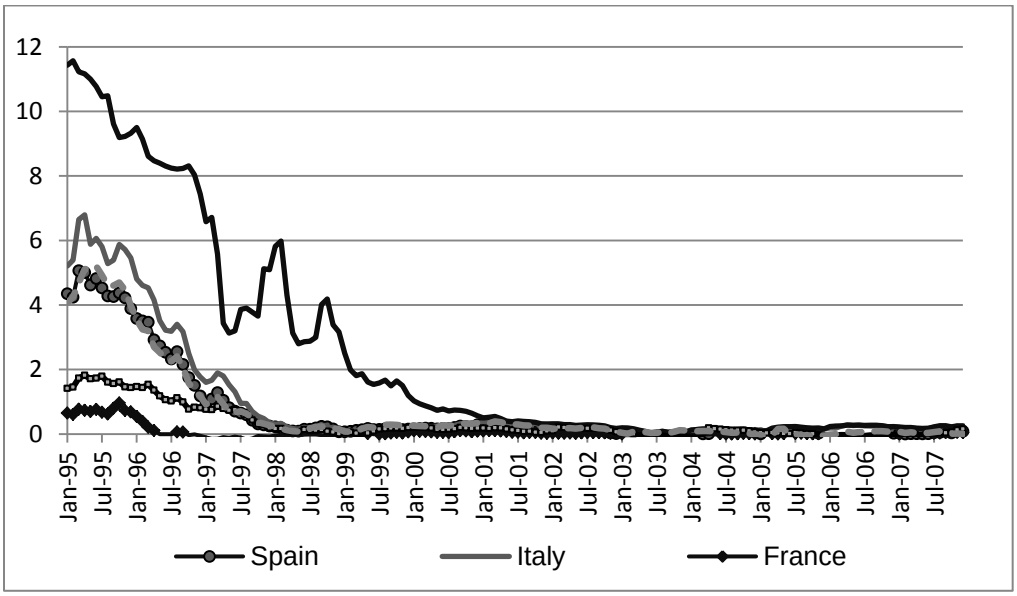
Note: Portugese unemployment data are estimated.

Source: Eurostat, "Unemployment rate, annual average, by sex and age groups (%)," [une_rt_a], updated April 11, 2013, accessed April 29, 2013.

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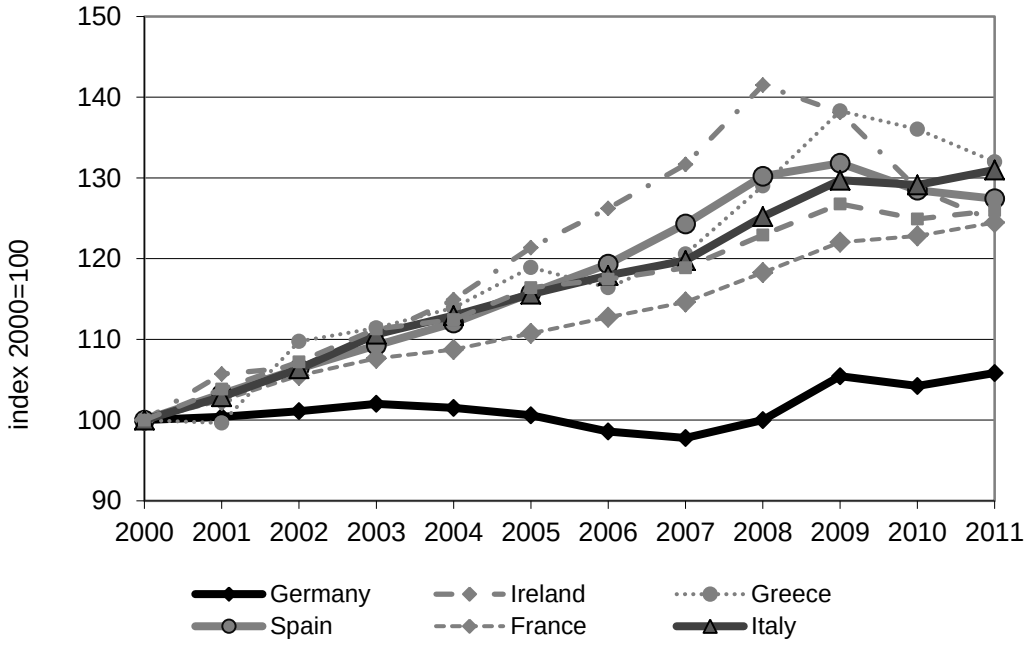
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Exhibit 11 Ten-year government bond spreads over Germany, 1995-2007



Source: Global Financial Data, accessed April 16, 2012, series IGDEU10D (Germany), IGESP10D (Spain), IGITA10D (Italy), IGFRA10D (France), IGGRC10D (Greece), IGIRL10D (Ireland) and IGPRT10D (Portugal).

Exhibit 12 Unit labor costs (euros per unit of output), selected Eurozone countries



Source: Eurostat, "Nominal unit labour cost, "nama_aux_ulc, Index, 2005=100, updated Feb. 13, 2102, rebased by authors to an index value of 2000 = 100.

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Exhibit 13 Dow Jones stock market indices 2011 through 2012 (index January 2011 = 0)

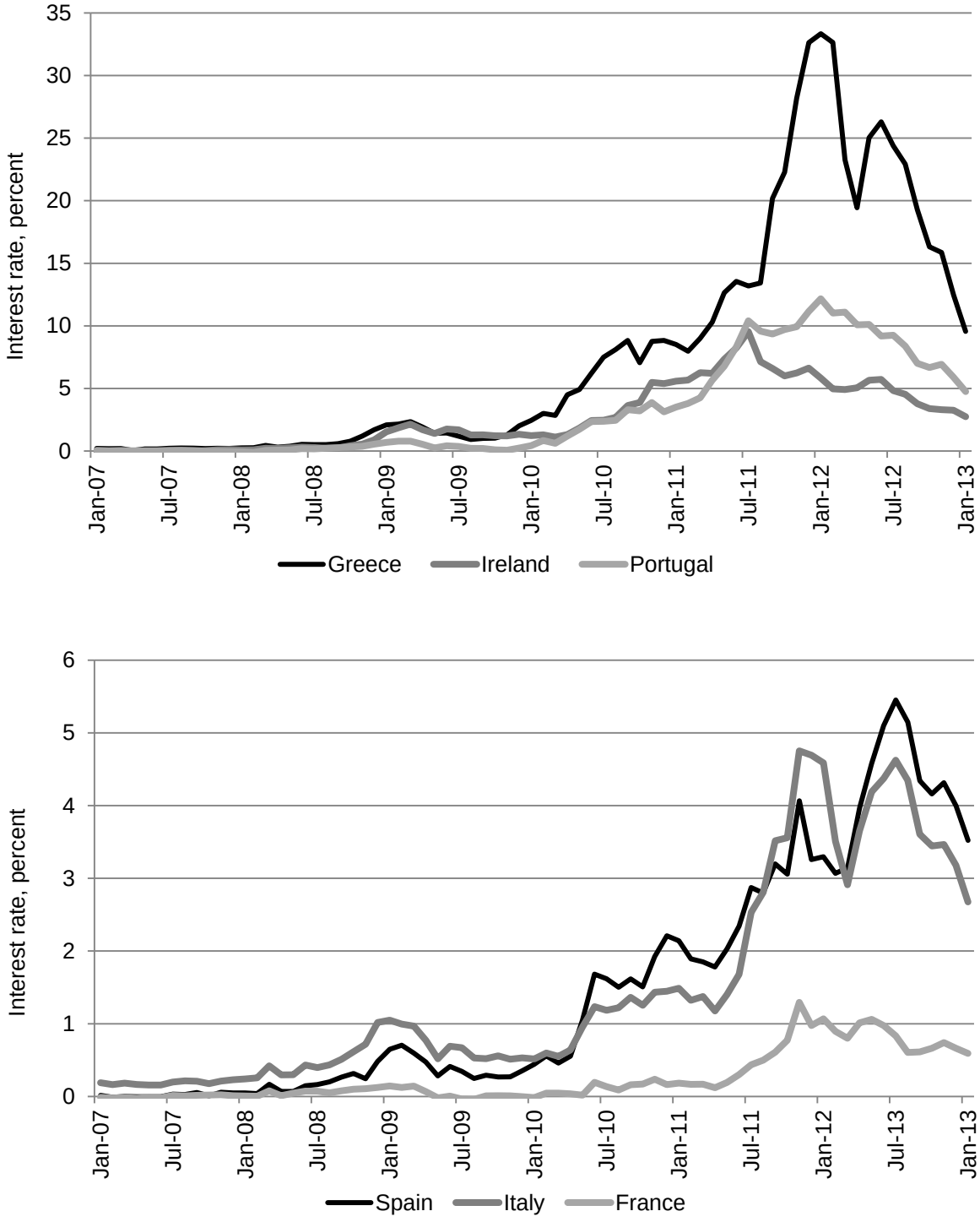


Source: Global Financial Data, accessed Apr. 5, 2013: Dow Jones Germany Stock Index, _DE1; Dow Jones Greece Stock Index, _GR1; Dow Jones France Stock Index, _FR1; Dow Jones Italy Stock Index, _IT1.

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Exhibit 14 Spreads of 10-year government bonds over Germany, 2007 through 2012

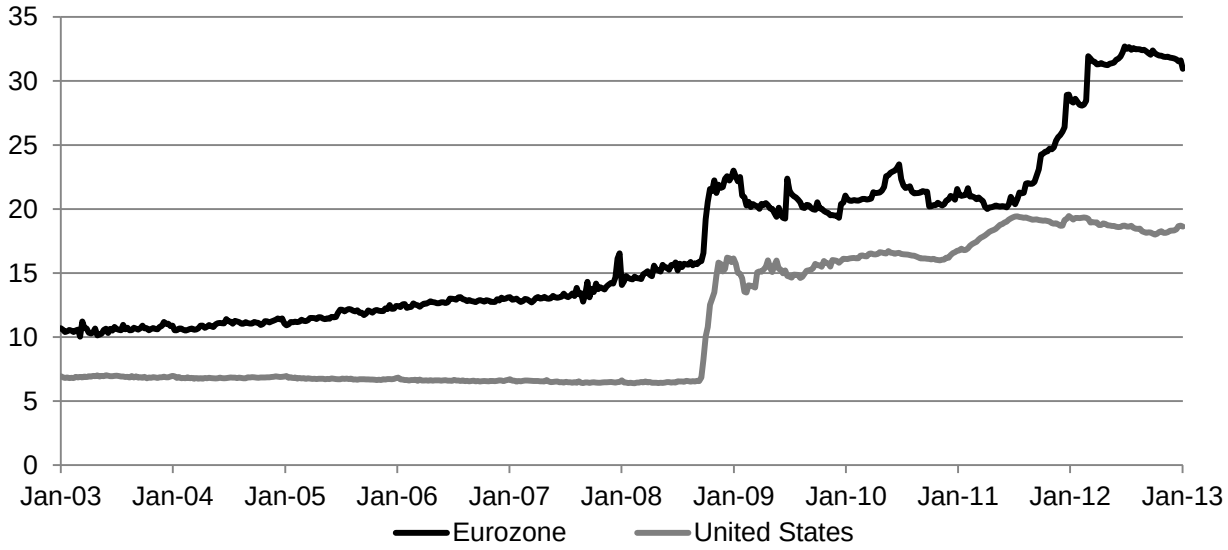


Source: Global Financial Data, accessed April 4, 2013, series IGDEU10D (Germany), IGESP10D (Spain), IGITA10D (Italy), IGFRA10D (France), IGGRC10D (Greece), IGIRL10D (Ireland) and IGPRT10D (Portugal).

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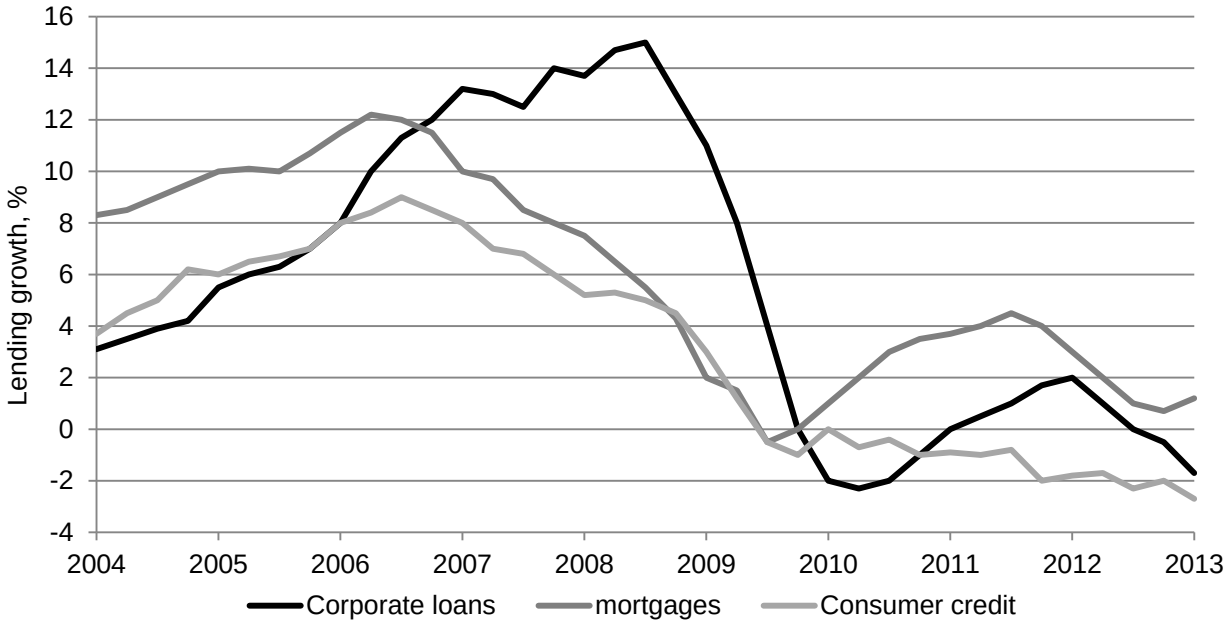
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Exhibit 15 Total central bank liabilities as % of GDP, 2003 through 2012



Sources: Federal Reserve Assets: Board of Governors of the Federal Reserve, H.4.1 Historical Data, Table 1, "Factors Affecting Reserve Balances," www.federalreserve.gov/datadownload/Choose.aspx?rel=H41, accessed March 31, 2013. U.S. GDP: Bureau of Economic Analysis, Table 1.1.5. Gross Domestic Product, seasonally adjusted (quarterly at annual rates), accessed April 30, 2013. ECB assetes: ECB Statistical Data Warehouse, Reports > Monthly Bulletin > 1 Monetary policy statistics > 1.1 Consolidated financial statement of the Eurosystem, http://sdw.ecb.europa.eu/browse.do?node=bbn129, accessed March 27, 2013. Euro 17 GDP: Eurostat, GDP and main components - Current prices [nama_gdp_c] , Gross domestic product at market prices, Euro area (17 countries), accessed April 30, 2013. For the United States and the Eurozone, weekly GDP was estimated from quarterly and annual data by linear interpolation.

Exhibit 16 Growth in lending by Eurozone banks



Source: Adapted from Morgan Stanley Research Europe, Banks/Rates/Credit , January 15, 2013

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Endnotes

[1] Global Financial Data, accessed April 16, 2012, series IGGRC10D (Greece).

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[10] Rawi Abdelal, "The Politics of Monetary Leadership and Followership: Stability in the European Monetary System Since the Currency Crisis of 1992." *Political Studies* 46, no. 2 (June 1998), pp. 246-47.

[11] Charles R. Bean, "Economic and Monetary Union in Europe," *Journal of Economic Perspectives*, 6:4 (fall 1992), summarizing widely held views, not stating his own opinion.

[12] Geoffrey Garrett, "The Politics of Maastricht," *Economics and Politics* 5: 2 (July 1993), 115-116. German unification occurred before Maastricht, but it would have been difficult for Kohl to go back on the deal.

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[15] Martin Feldstein, "EMU and International Conflict."

[16] Norman Lamont, *In Office*, London: Little, Brown, and Co., 1999, p. 118-119.

[17] Tony Barber, "Saving the euro: Tall ambition, flawed foundations," Financial Times, October 11, 2010.

[18] Geert De Clercq, "Reuters Special Report: Architects of a currency in crisis," Reuters, Dec 30, 2011.

[19] Martin Feldstein, "EMU and International Conflict," p. 61.

[20] "Consolidated Version of the Treaty on the Functioning of the European Union," *Official Journal of the European Union,* March 30, 2010, from Europa (official EU website), http://eur-lex.europa.eu/, accessed December 9, 2011.

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[27] Jean Pisani-Ferry, "The Euro Crisis and the New Impossible Trinity."

[28] Merrill Lynch Private Banking and Investment Group, "Waiting for the end," CIO Outlook, September 2011, gives this well-known data.

[29] Discussion of the banking sector draws from Jean Pisani-Ferry, "The Euro Crisis and the New Impossible Trinity."

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[39] Deeg, p. 340.

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